

William Stallings

Data and Computer

Communications

Chapter 13

Local Area Network

Technology

LAN Applications (1)

- Personal computer LANs
 - Low cost
 - Limited data rate
- Back end networks and storage area networks
 - Interconnecting large systems (mainframes and large storage devices)
 - High data rate
 - High speed interface
 - Distributed access
 - Limited distance
 - Limited number of devices

LAN Applications (2)

- High speed office networks
 - Desktop image processing
 - High capacity local storage
- Backbone LANs
 - Interconnect low speed local LANs
 - Reliability
 - Capacity
 - Cost

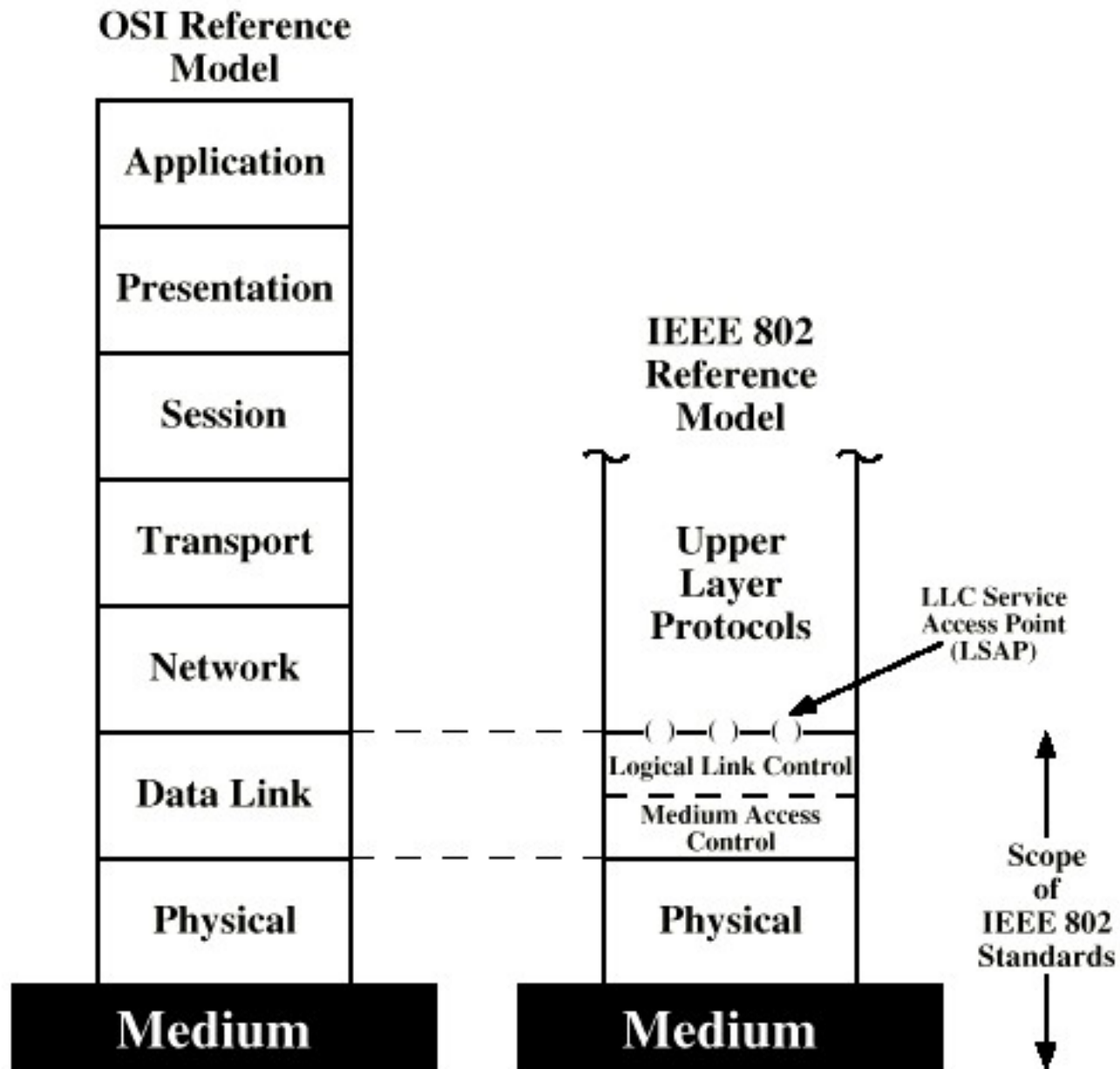
LAN Architecture

- Protocol architecture
- Topologies
- Media access control
- Logical Link Control

Protocol Architecture

- Lower layers of OSI model
- IEEE 802 reference model
- Physical
- Logical link control (LLC)
- Media access control (MAC)

IEEE 802 v OSI



802 Layers - Physical

- Encoding/decoding
- Preamble generation/removal
- Bit transmission/reception
- Transmission medium and topology

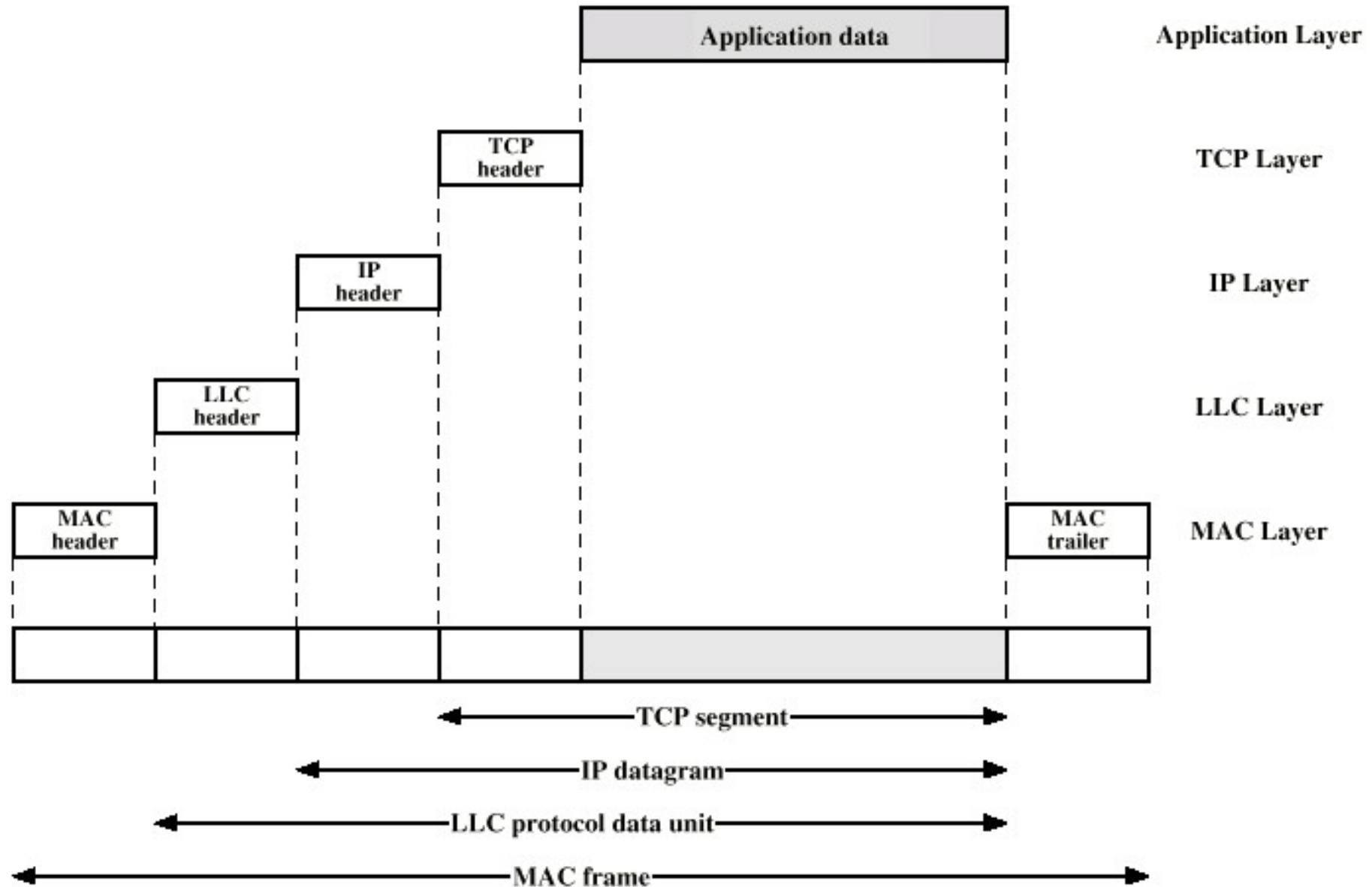
802 Layers - Logical Link Control

- Interface to higher levels
- Flow and error control

802 Layers - Media Access Control

- Assembly of data into frame with address and error detection fields
- Disassembly of frame
 - Address recognition
 - Error detection
- Govern access to transmission medium
 - Not found in traditional layer 2 data link control
- For the same LLC, several MAC options may be available

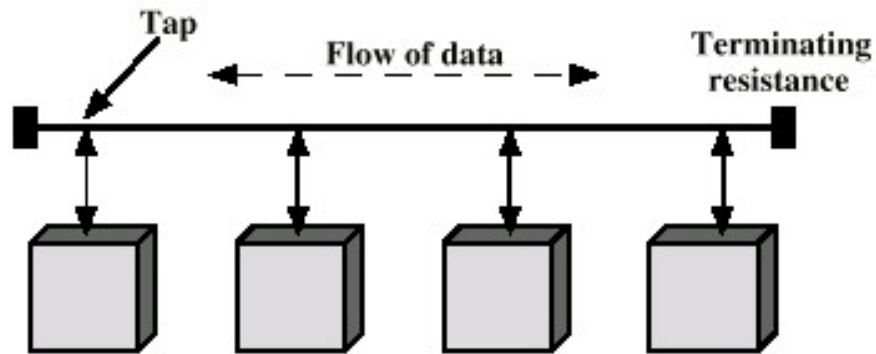
LAN Protocols in Context



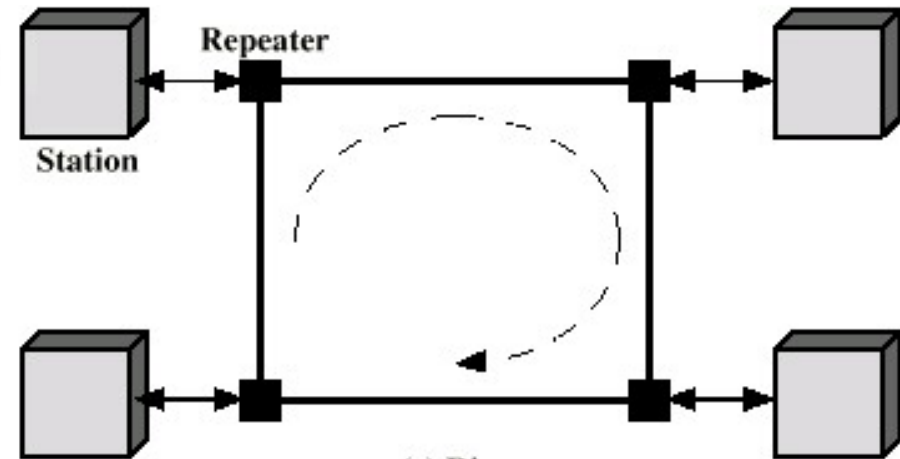
Topologies

- Tree
- Bus
 - Special case of tree
 - One trunk, no branches
- Ring
- Star

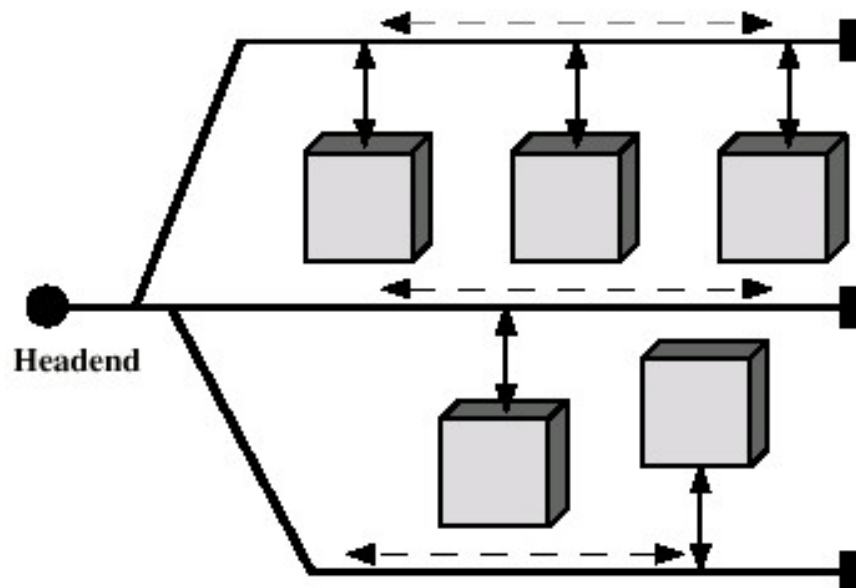
LAN Topologies



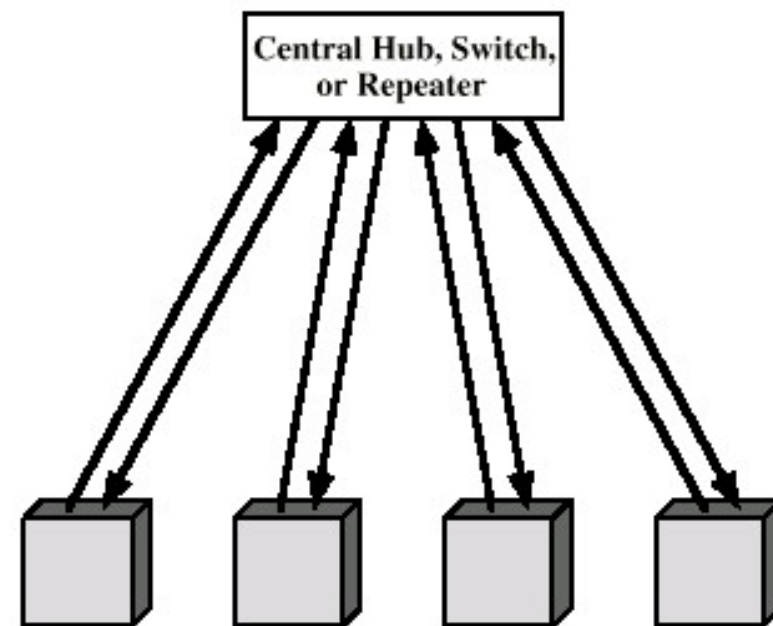
(a) Bus



(c) Ring



(b) Tree

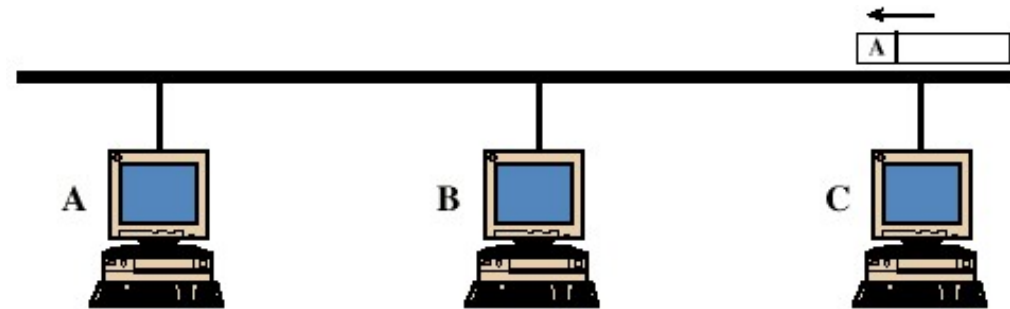


(d) Star

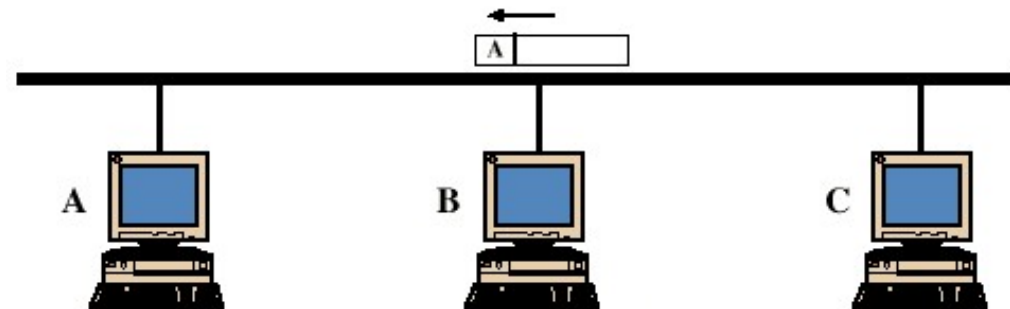
Bus and Tree

- Multipoint medium
- Transmission propagates throughout medium
- Heard by all stations
 - Need to identify target station
 - Each station has unique address
- Full duplex connection between station and tap
 - Allows for transmission and reception
- Need to regulate transmission
 - To avoid collisions
 - To avoid hogging
 - Data in small blocks - frames
- Terminator absorbs frames at end of medium

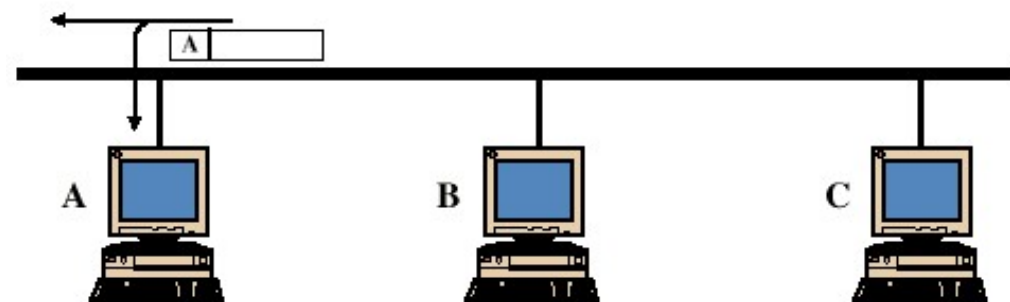
Frame Transmission - Bus LAN



C transmits frame addressed to A



Frame is not addressed to B; B ignores it



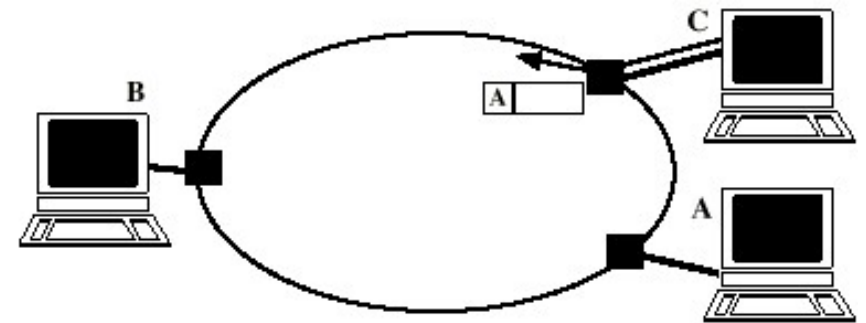
A copies frame as it goes by

Ring Topology

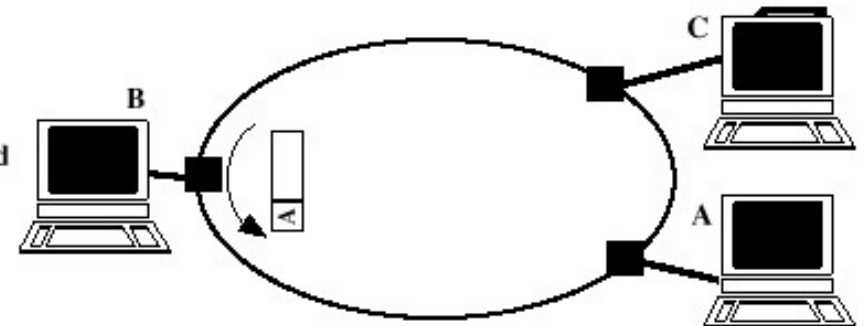
- Repeaters joined by point to point links in closed loop
 - Receive data on one link and retransmit on another
 - Links unidirectional
 - Stations attach to repeaters
- Data in frames
 - Circulate past all stations
 - Destination recognizes address and copies frame
 - Frame circulates back to source where it is removed
- Media access control determines when station can insert frame

Frame Transmission Ring LAN

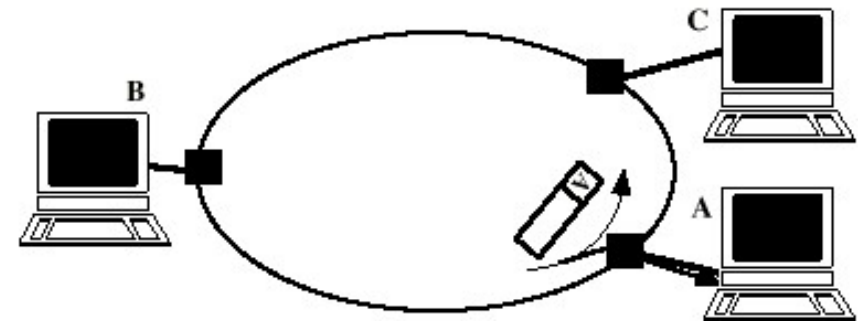
(a) C transmits frame addressed to A



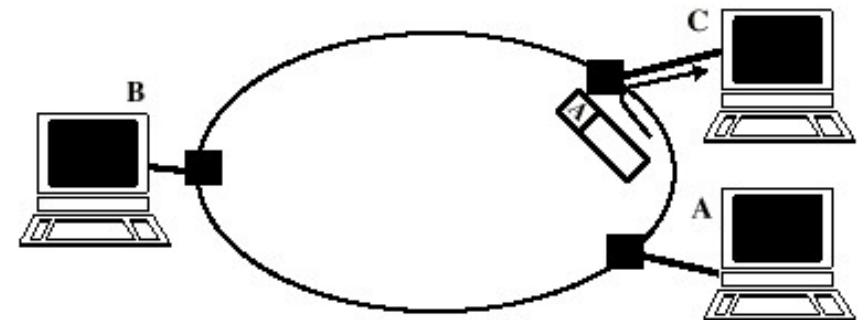
(b) Frame is not addressed to B; B ignores it



(c) A copies frame as it goes by



(d) C absorbs returning frame



Star Topology

- Each station connected directly to central node
 - Usually via two point to point links
- Central node can broadcast
 - Physical star, logical bus
 - Only one station can transmit at a time
- Central node can act as frame switch

Media Access Control

□ Where

□ Central

- Greater control
- Simple access logic at station
- Avoids problems of co-ordination
- Single point of failure
- Potential bottleneck

□ Distributed

□ How

□ Synchronous

- Specific capacity dedicated to connection

□ Asynchronous

- In response to demand

Asynchronous Systems

- Round robin
 - Good if many stations have data to transmit over extended period
- Reservation
 - Good for stream traffic
- Contention
 - Good for bursty traffic
 - All stations contend for time
 - Distributed
 - Simple to implement
 - Efficient under moderate load
 - Tend to collapse under heavy load

MAC Frame Format

- ❑ MAC layer receives data from LLC layer
- ❑ MAC control
- ❑ Destination MAC address
- ❑ Source MAC address
- ❑ LLS
- ❑ CRC
- ❑ MAC layer detects errors and discards frames
- ❑ LLC optionally retransmits unsuccessful frames

Logical Link Control

- Transmission of link level PDUs between two stations
- Must support multiaccess, shared medium
- Relieved of some link access details by MAC layer
- Addressing involves specifying source and destination LLC users
 - Referred to as service access points (SAP)
 - Typically higher level protocol

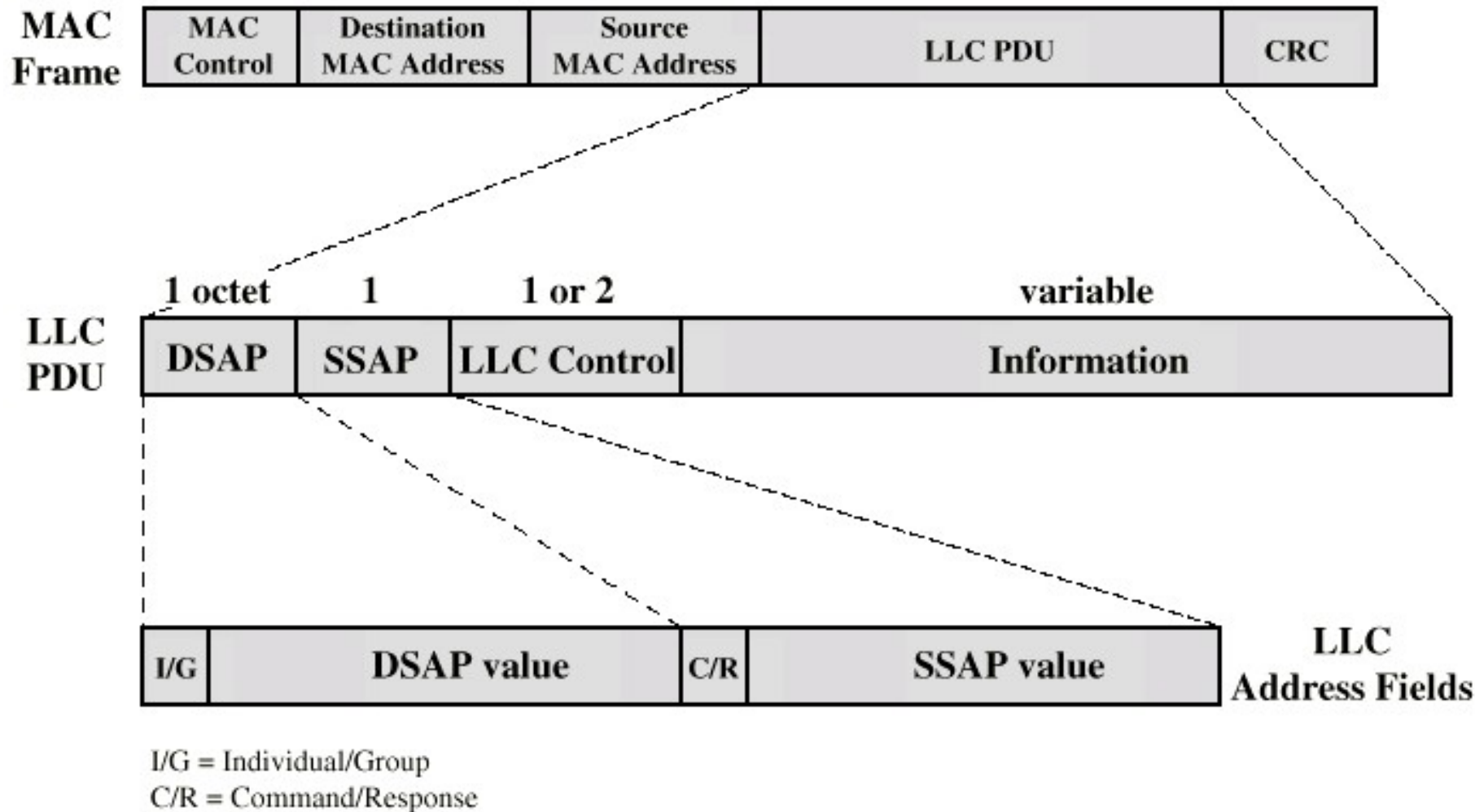
LLC Services

- Based on HDLC
- Unacknowledged connectionless service
- Connection mode service
- Acknowledged connectionless service

LLC Protocol

- Modeled after HDLC
- Asynchronous balanced mode to support connection mode LLC service (type 2 operation)
- Unnumbered information PDUs to support Acknowledged connectionless service (type 1)
- Multiplexing using LSAPs

Typical Frame Format



Bus LANs

- Signal balancing
 - Signal must be strong enough to meet receiver's minimum signal strength requirements
 - Give adequate signal to noise ration
 - Not so strong that it overloads transmitter
 - Must satisfy these for all combinations of sending and receiving station on bus
 - Usual to divide network into small segments
 - Link segments with amplifies or repeaters

Transmission Media

- Twisted pair
 - Not practical in shared bus at higher data rates
- Baseband coaxial cable
 - Used by Ethernet
- Broadband coaxial cable
 - Included in 802.3 specification but no longer made
- Optical fiber
 - Expensive
 - Difficulty with availability
 - Not used
- Few new installations
 - Replaced by star based twisted pair and optical fiber

Baseband Coaxial Cable

- Uses digital signaling
- Manchester or Differential Manchester encoding
- Entire frequency spectrum of cable used
- Single channel on cable
- Bi-directional
- Few kilometer range
- Ethernet (basis for 802.3) at 10Mbps
- 50 ohm cable

10Base5

- Ethernet and 802.3 originally used 0.4 inch diameter cable at 10Mbps
- Max cable length 500m
- Distance between taps a multiple of 2.5m
 - Ensures that reflections from taps do not add in phase
- Max 100 taps
- 10Base5

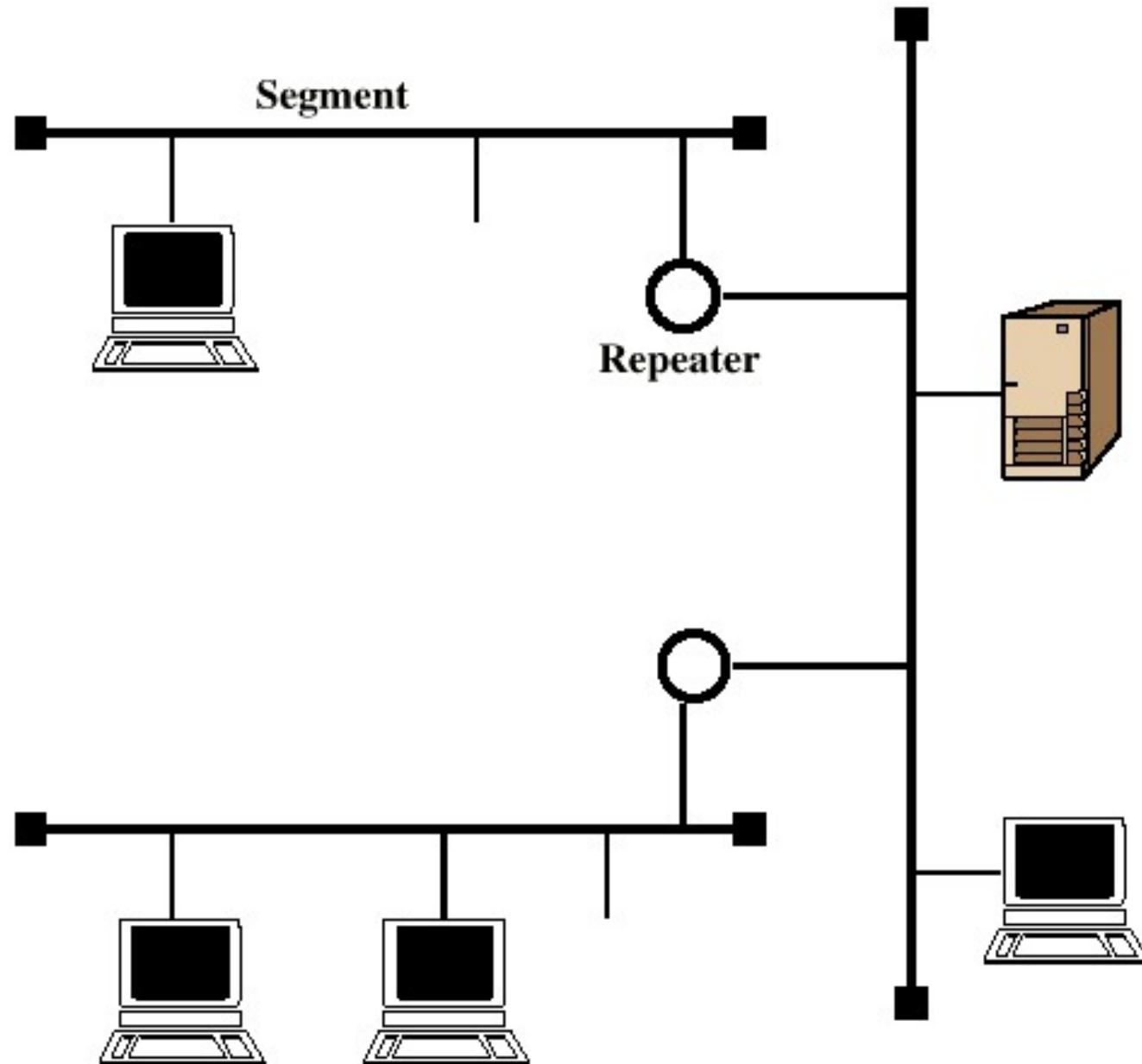
10Base2

- ❑ Cheaper
- ❑ 0.25 inch cable
 - ❑ More flexible
 - ❑ Easier to bring to workstation
 - ❑ Cheaper electronics
 - ❑ Greater attenuation
 - ❑ Lower noise resistance
 - ❑ Fewer taps (30)
 - ❑ Shorter distance (185m)

Repeaters

- Transmits in both directions
- Joins two segments of cable
- No buffering
- No logical isolation of segments
- If two stations on different segments send at the same time, packets will collide
- Only one path of segments and repeaters between any two stations

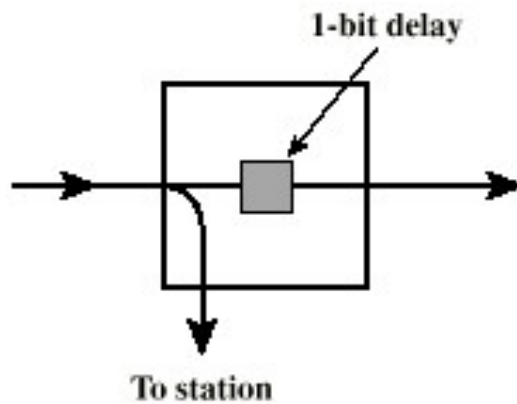
Baseband Configuration



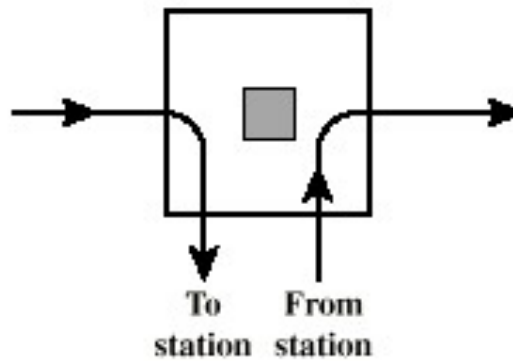
Ring LANs

- Each repeater connects to two others via unidirectional transmission links
- Single closed path
- Data transferred bit by bit from one repeater to the next
- Repeater regenerates and retransmits each bit
- Repeater performs data insertion, data reception, data removal
- Repeater acts as attachment point
- Packet removed by transmitter after one trip round ring

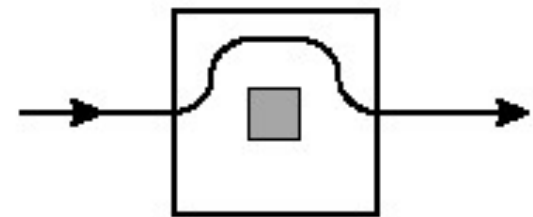
Ring Repeater States



(a) Listen state



(b) Transmit state



(c) Bypass state

Listen State Functions

- Scan passing bit stream for patterns
 - Address of attached station
 - Token permission to transmit
- Copy incoming bit and send to attached station
 - Whilst forwarding each bit
- Modify bit as it passes
 - e.g. to indicate a packet has been copied (ACK)

Transmit State Functions

- Station has data
- Repeater has permission
- May receive incoming bits
 - If ring bit length shorter than packet
 - Pass back to station for checking (ACK)
 - May be more than one packet on ring
 - Buffer for retransmission later

Bypass State

- Signals propagate past repeater with no delay (other than propagation delay)
- Partial solution to reliability problem (see later)
- Improved performance

Ring Media

- Twisted pair
- Baseband coaxial
- Fiber optic
- Not broadband coaxial
 - Would have to receive and transmit on multiple channels, asynchronously

Timing Jitter

- Clocking included with signal
 - e.g. differential Manchester encoding
 - Clock recovered by repeaters
 - To know when to sample signal and recover bits
 - Use clocking for retransmission
 - Clock recovery deviates from midbit transmission randomly
 - Noise
 - Imperfections in circuitry
- Retransmission without distortion but with timing error
- Cumulative effect is that bit length varies
- Limits number of repeaters on ring

Solving Timing Jitter Limitations

- Repeater uses phase locked loop
 - Minimize deviation from one bit to the next
- Use buffer at one or more repeaters
 - Hold a certain number of bits
 - Expand and contract to keep bit length of ring constant
- Significant increase in maximum ring size

Potential Ring Problems

- Break in any link disables network
- Repeater failure disables network
- Installation of new repeater to attach new station requires identification of two topologically adjacent repeaters
- Timing jitter
- Method of removing circulating packets required
 - With backup in case of errors
- Mostly solved with star-ring architecture

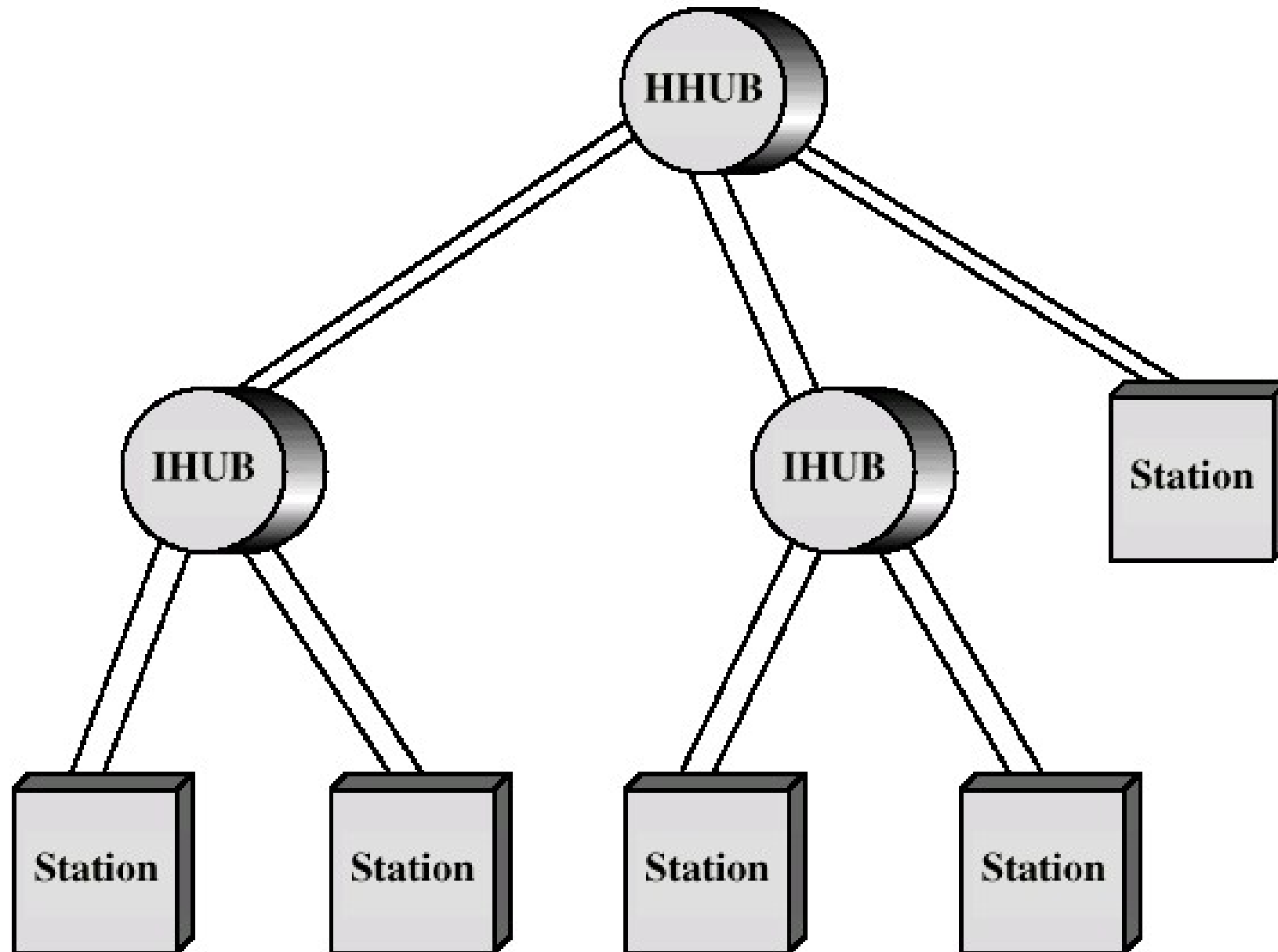
Star Ring Architecture

- Feed all inter-repeater links to single site
 - Concentrator
 - Provides central access to signal on every link
 - Easier to find faults
 - Can launch message into ring and see how far it gets
 - Faulty segment can be disconnected and repaired later
 - New repeater can be added easily
 - Bypass relay can be moved to concentrator
 - Can lead to long cable runs
- Can connect multiple rings using bridges

Star LANs

- Use unshielded twisted pair wire (telephone)
 - Minimal installation cost
 - May already be an installed base
 - All locations in building covered by existing installation
- Attach to a central active hub
- Two links
 - Transmit and receive
- Hub repeats incoming signal on all outgoing lines
- Link lengths limited to about 100m
 - Fiber optic - up to 500m
- Logical bus - with collisions

Two Level Star Topology



Hubs and Switches

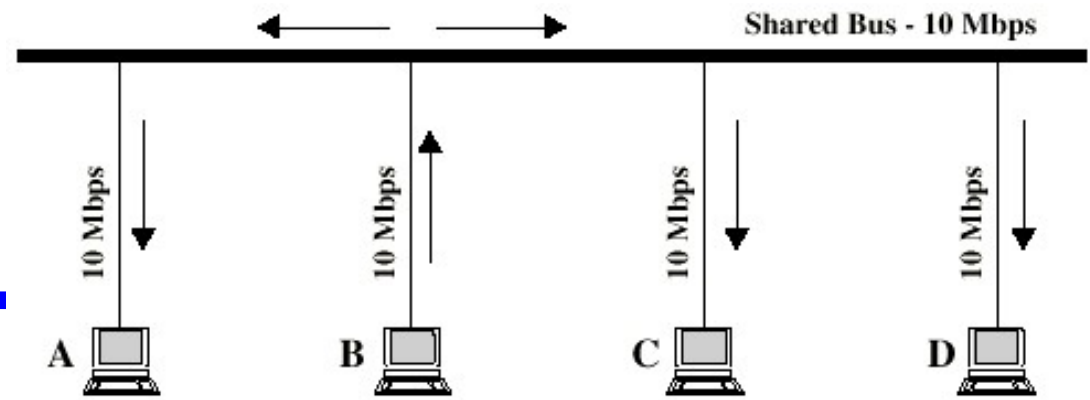
- Shared medium hub
 - Central hub
 - Hub retransmits incoming signal to all outgoing lines
 - Only one station can transmit at a time
 - With a 10Mbps LAN, total capacity is 10Mbps
- Switched LAN hub
 - Hub acts as switch
 - Incoming frame switches to appropriate outgoing line
 - Unused lines can also be used to switch other traffic
 - With two pairs of lines in use, overall capacity is now 20Mbps

Switched Hubs

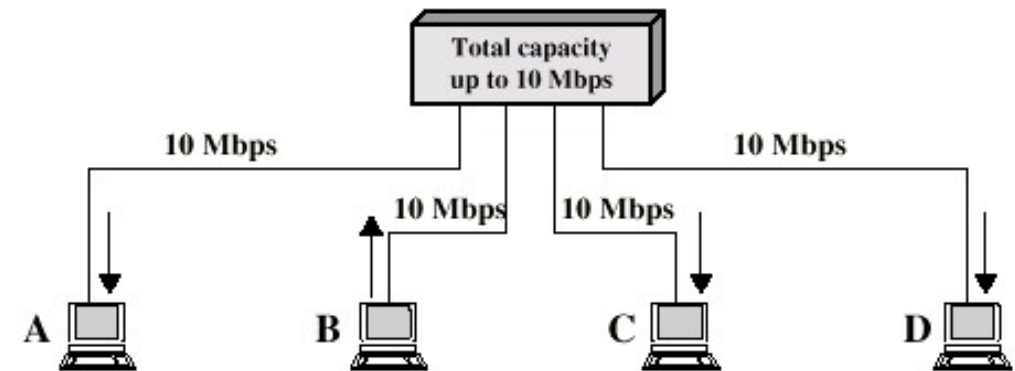
- No change to software or hardware of devices
- Each device has dedicated capacity
- Scales well

- Store and forward switch
 - Accept input, buffer it briefly, then output
- Cut through switch
 - Take advantage of the destination address being at the start of the frame
 - Begin repeating incoming frame onto output line as soon as address recognized
 - May propagate some bad frames

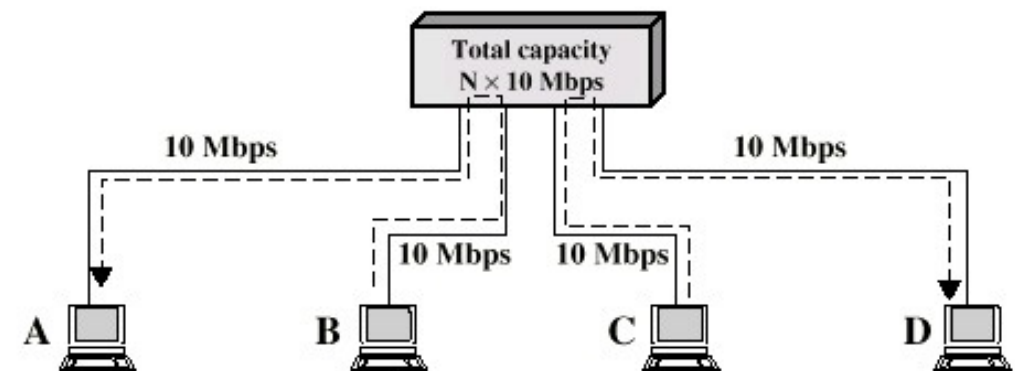
Hubs and Switches (diag)



(a) Shared medium bus



(b) Shared medium hub



(c) Switching hub

Wireless LANs

- Mobility
- Flexibility
- Hard to wire areas
- Reduced cost of wireless systems
- Improved performance of wireless systems

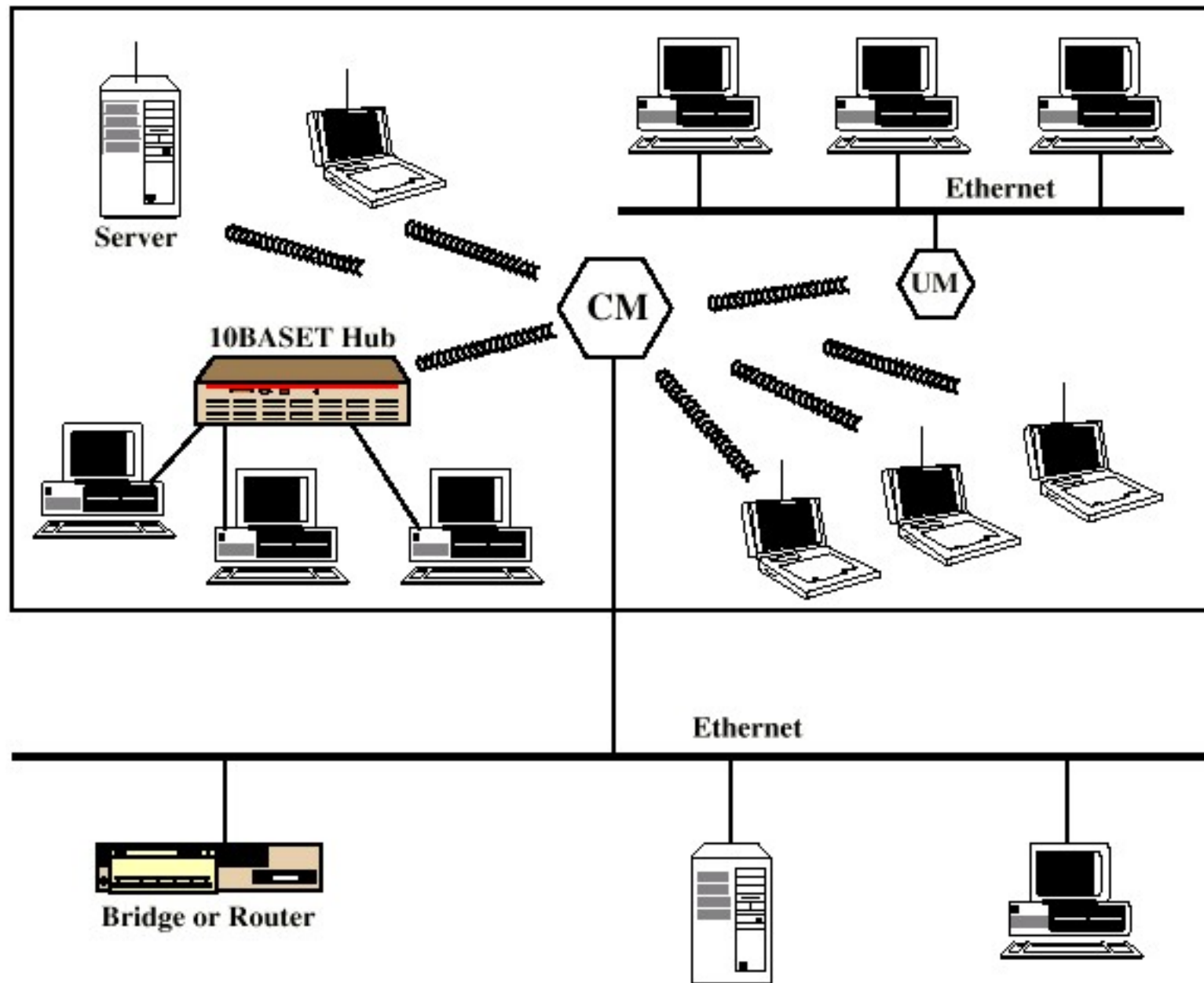
Wireless LAN Applications

- LAN Extension
- Cross building interconnection
- Nomadic access
- Ad hoc networks

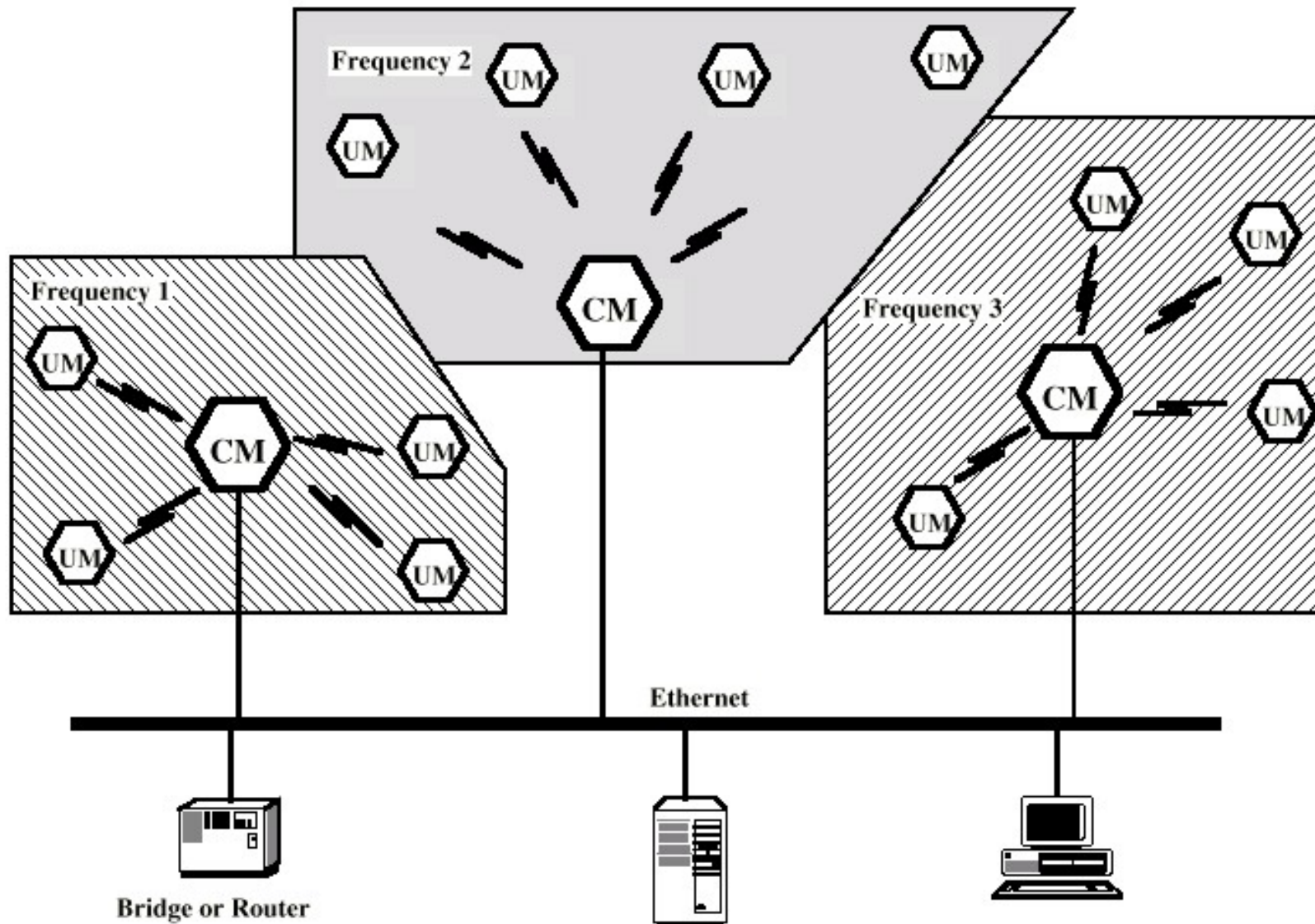
LAN Extension

- Buildings with large open areas
 - Manufacturing plants
 - Warehouses
- Historical buildings
- Small offices
- May be mixed with fixed wiring system

Single Cell Wireless LAN



Multi Cell Wireless LAN



Cross Building Interconnection

- Point to point wireless link between buildings
- Typically connecting bridges or routers
- Used where cable connection not possible
 - e.g. across a street

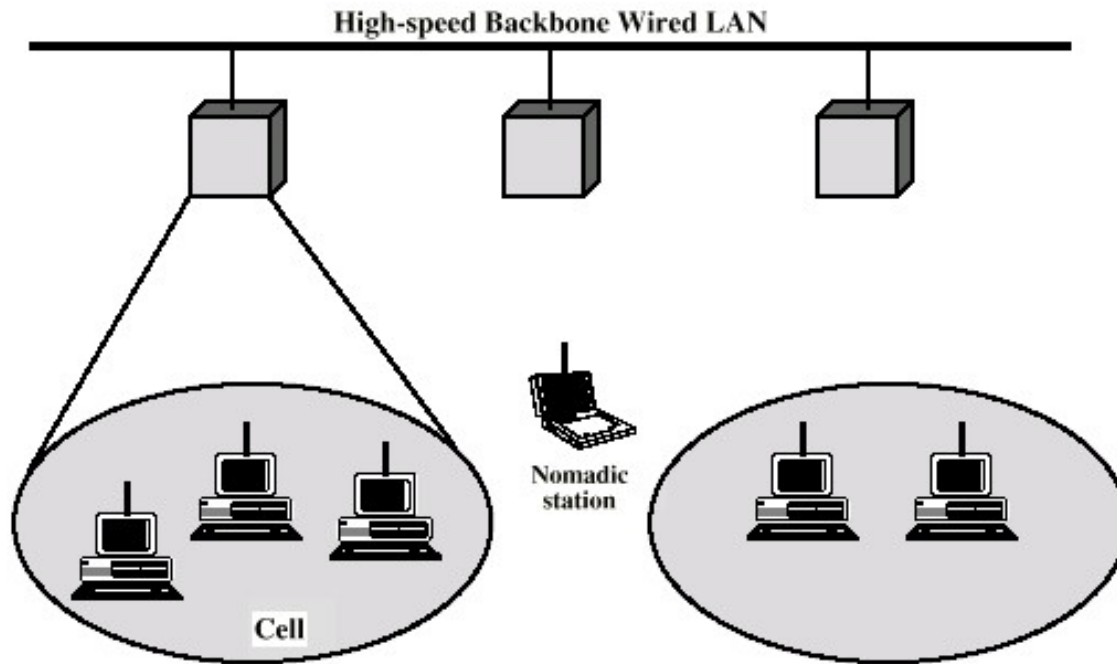
Nomadic Access

- Mobile data terminal
 - e.g. laptop
- Transfer of data from laptop to server
- Campus or cluster of buildings

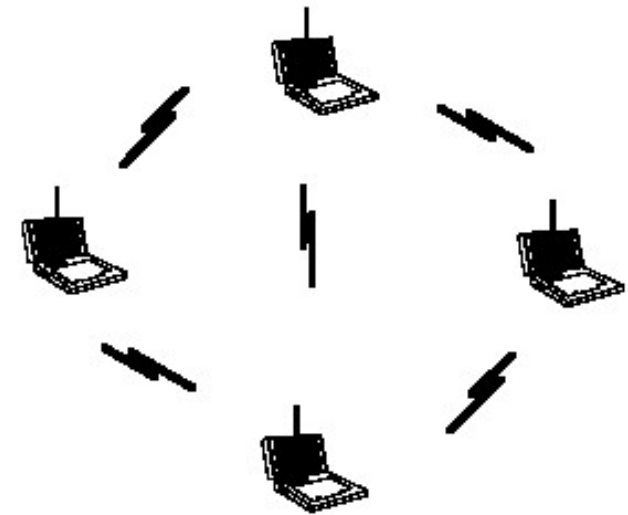
Ad Hoc Networking

- Peer to peer
- Temporary
- e.g. conference

Wireless LAN Configurations



(a) Infrastructure Wireless LAN



(b) Ad hoc LAN

Wireless LAN Requirements

- Throughput
- Number of nodes
- Connection to backbone
- Service area
- Battery power consumption
- Transmission robustness and security
- Collocated network operation
- License free operation
- Handoff/roaming
- Dynamic configuration

Wireless LAN Technology

- Infrared (IR) LANs
- Spread spectrum LANs
- Narrow band microwave

Bridges

- Ability to expand beyond single LAN
- Provide interconnection to other LANs/WANs
- Use Bridge or router
- Bridge is simpler
 - Connects similar LANs
 - Identical protocols for physical and link layers
 - Minimal processing
- Router more general purpose
 - Interconnect various LANs and WANs
 - see later

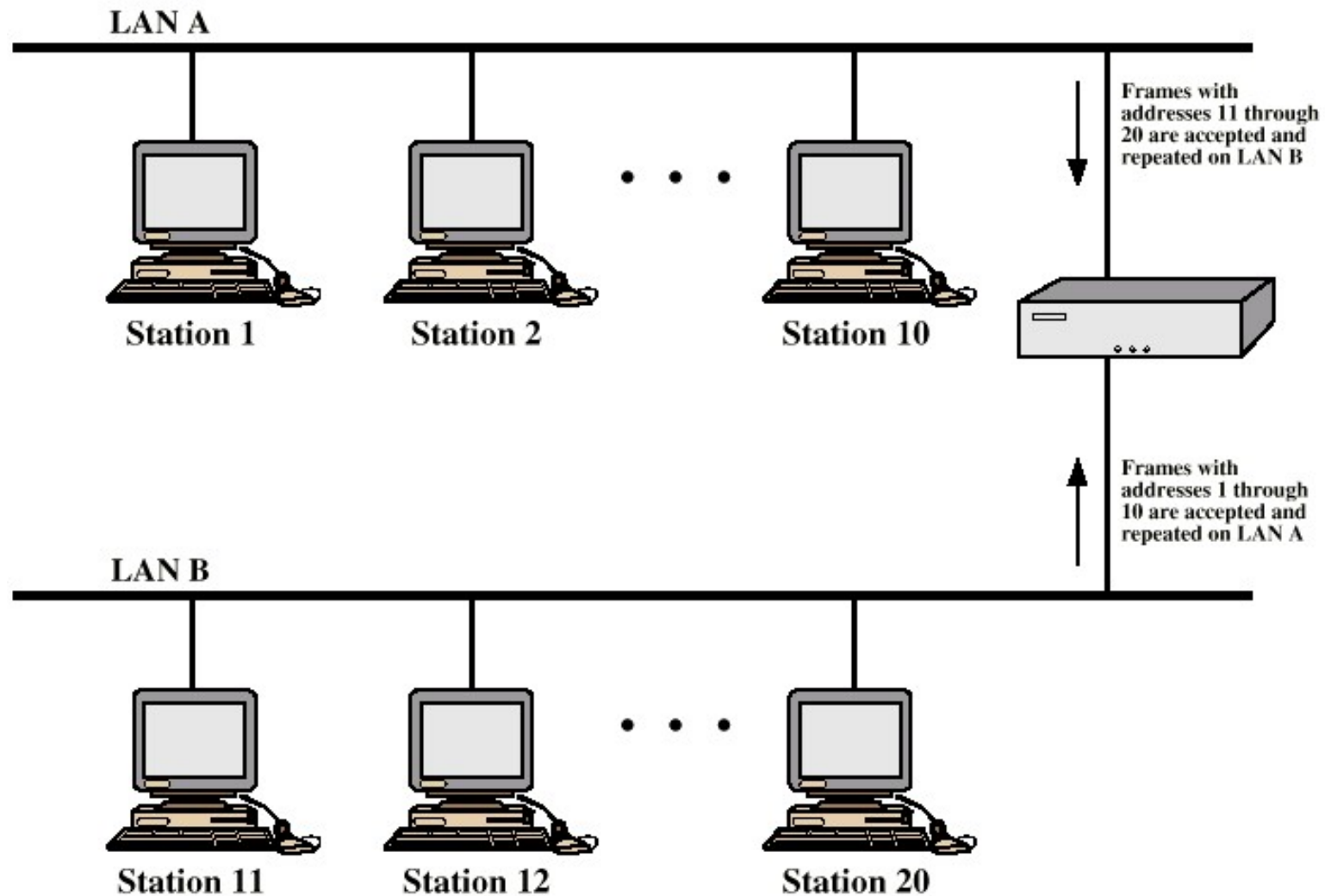
Why Bridge?

- Reliability
- Performance
- Security
- Geography

Functions of a Bridge

- Read all frames transmitted on one LAN and accept those address to any station on the other LAN
- Using MAC protocol for second LAN, retransmit each frame
- Do the same the other way round

Bridge Operation



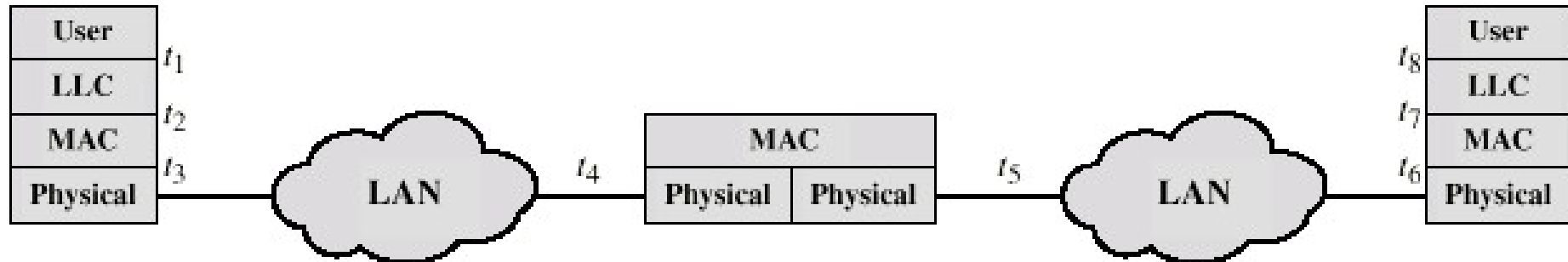
Bridge Design Aspects

- No modification to content or format of frame
- No encapsulation
- Exact bitwise copy of frame
- Minimal buffering to meet peak demand
- Contains routing and address intelligence
 - Must be able to tell which frames to pass
 - May be more than one bridge to cross
- May connect more than two LANs
- Bridging is transparent to stations
 - Appears to all stations on multiple LANs as if they are on one single LAN

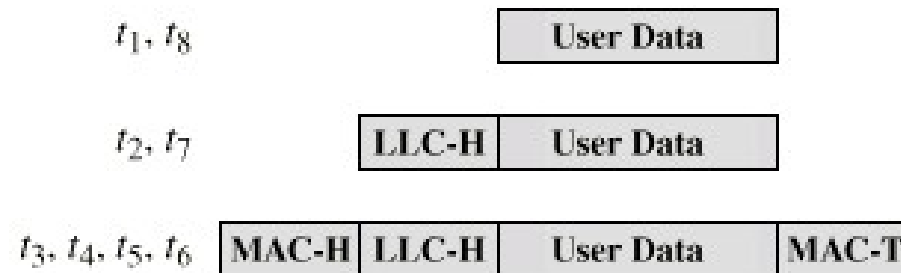
Bridge Protocol Architecture

- IEEE 802.1D
- MAC level
 - Station address is at this level
- Bridge does not need LLC layer
 - It is relaying MAC frames
- Can pass frame over external comms system
 - e.g. WAN link
 - Capture frame
 - Encapsulate it
 - Forward it across link
 - Remove encapsulation and forward over LAN link

Connection of Two LANs



(a) Architecture

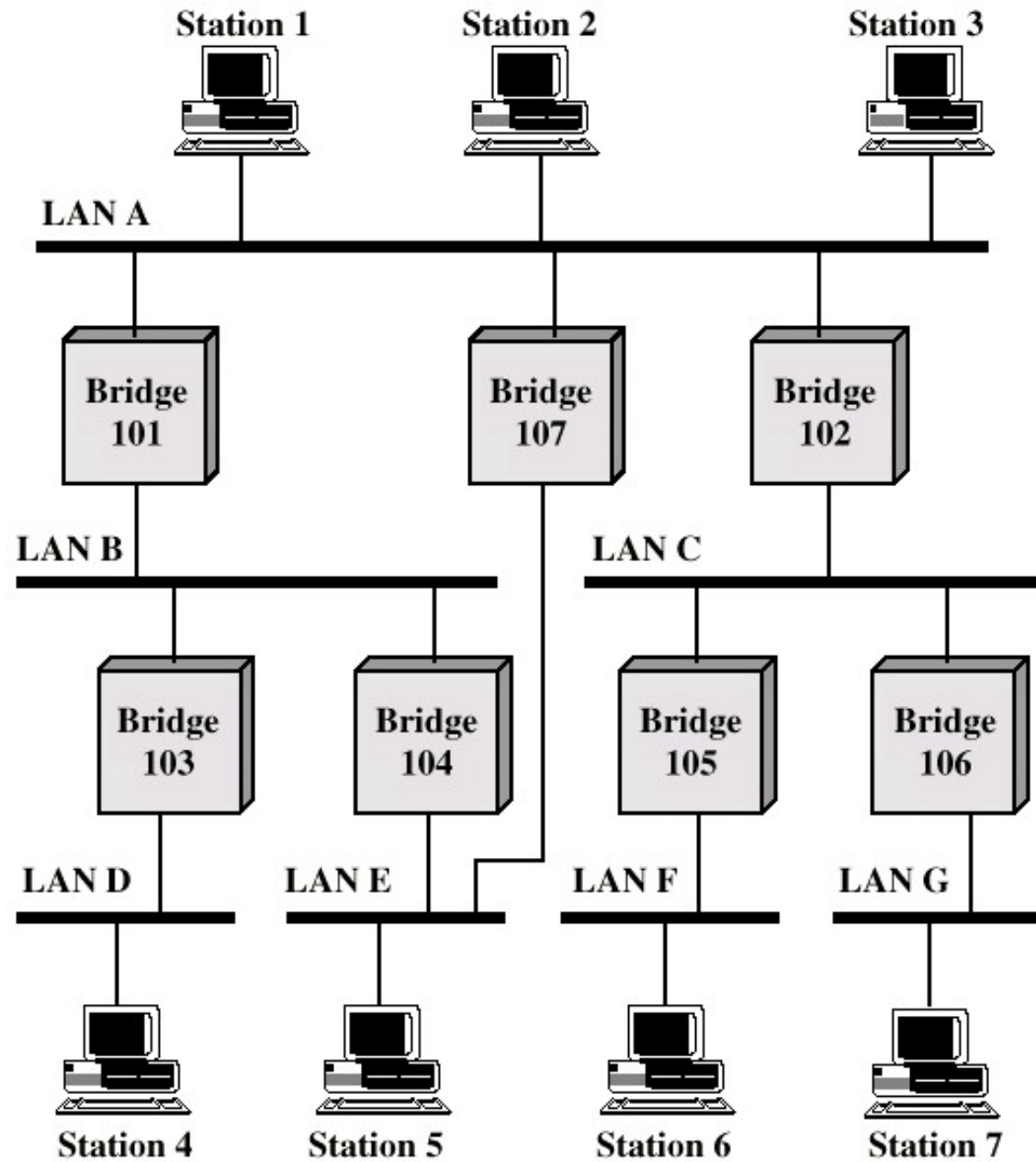


(b) Operation

Fixed Routing

- Complex large LANs need alternative routes
 - Load balancing
 - Fault tolerance
- Bridge must decide whether to forward frame
- Bridge must decide which LAN to forward frame on
- Routing selected for each source-destination pair of LANs
 - Done in configuration
 - Usually least hop route
 - Only changed when topology changes

Multiple LANs



Spanning Tree

- Bridge automatically develops routing table
- Automatically update in response to changes
- Frame forwarding
- Address learning
- Loop resolution

Frame forwarding

- Maintain forwarding database for each port
 - List station addresses reached through each port
- For a frame arriving on port X:
 - Search forwarding database to see if MAC address is listed for any port except X
 - If address not found, forward to all ports except X
 - If address listed for port Y, check port Y for blocking or forwarding state
 - Blocking prevents port from receiving or transmitting
 - If not blocked, transmit frame through port Y

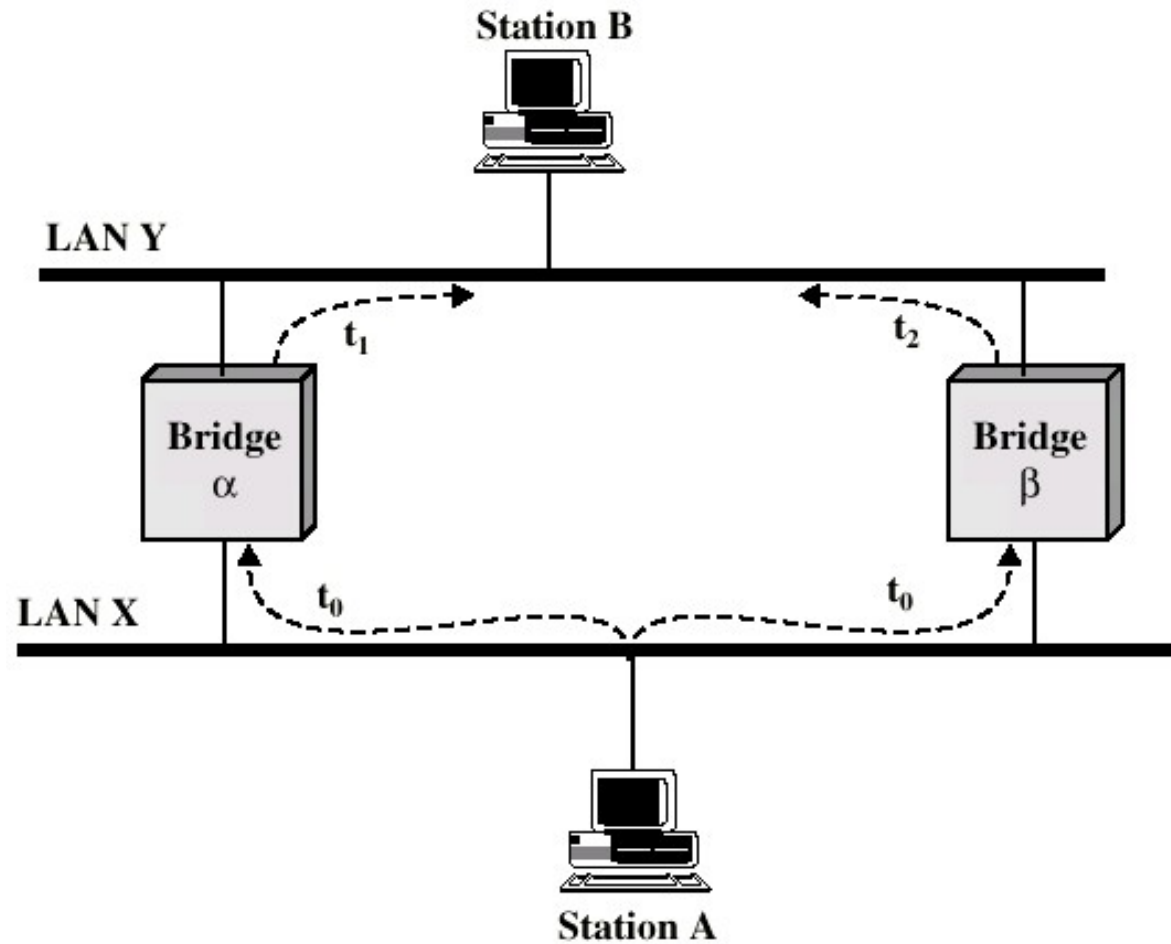
Address Learning

- Can preload forwarding database
- Can be learned
- When frame arrives at port X, it has come from the LAN attached to port X
- Use the source address to update forwarding database for port X to include that address
- Timer on each entry in database
- Each time frame arrives, source address checked against forwarding database

Spanning Tree Algorithm

- Address learning works for tree layout
 - i.e. no closed loops
- For any connected graph there is a spanning tree that maintains connectivity but contains no closed loops
- Each bridge assigned unique identifier
- Exchange between bridges to establish spanning tree

Loop of Bridges



Required Reading

- Stallings chapter 13
- Loads of info on the Web